Printed Pages: 2 **Sub Code: RPH-522**

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B. PHARM
(SEM V) THEORY EXAMINATION 2022-23
PHARMACEUTICS-VI (PHARMACEUTICAL TECHNOLOGY-I)

Time: 3 Hours ***Total Marks: 70***

Note: Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. 2*7 = 14

a. What is pre-formulation study?
b. Define the term "nano-emulsion".
c. Give examples of two preservatives used in biphasic liquid dosage forms.
d. What is skin permeation enhancer? Give one example.
e. What is an in-situ gel?
f. What are otic preparations?
g. State an advantage and a disadvantage of chloro fluoro carbons as propellent for pharmaceutical aerosols.

SECTION B

2. Attempt any *three* of the following: 7*3 = 21

a. Give brief introduction to ICH guidelines for stability of pharmaceuticals.
b. Write a short note on multiple emulsions and microemulsions.
c. Write a short note on the differentmanufacturing procedures and methods for the evaluationof suppositories.
d. Discuss in detail about the formulation of an ophthalmic product.
e. Discuss about the different evaluation parameters of a pharmaceutical aerosol.

SECTION C

3. Attempt any *one* part of the following: 7*1 = 7

a. Write a short note about the methods used for bulk characterization of drugs and discuss the importance of Preformulation studies.

b. State the importance of the dielectric constant and solubility analysis in formulation development.

4. Attempt any *one* part of the following: 7*1 = 7

a. Discuss the roles of vehicles, additives, stabilizers, preservatives and suspending agents in the formulation of a suspension with example.

b. Write a short note on the different stability problems that one can encounter while formulating an emulsion and discuss about the stabilization of a pharmaceutical emulsion.

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5. Attempt any *one* part of the following: 7*1 = 7

a. Write a detailed note on the different semisolid bases with proper examples.
b. Write the advantages and disadvantages of suppositories. Discuss the ideal features required for an ideal suppository base.

6. Attempt any *one* part of the following: 7*1 = 7

a. Write in detail about the large-scale manufacturing techniques of parenteral products.

b. Write in detail about the pre-filling treatments required in manufacturing of a parenteral preparation.

7. Attempt any *one* part of the following: 7*1 = 7

a. Write a short note on the general formulation of a pharmaceutical aerosol including various types of propellants used.

b. Write a short note on the regulatory requirements for approval of animal drugs including regulatory bodies and their guidelines.

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